Busy Beaver, Analysed

How one undecided 6-state machine became a problem in ergodic theory

BB(6) research program · 2026-09-08

1 The machine that won't sit still

The Busy Beaver number $\Sigma(6)$ is unknown, blocked by a few **holdout** Turing machines whose halting is equivalent to an open arithmetic conjecture. One holdout, watched on its natural macroscopic rhythm, is governed by an explicit integer map $\mathbf{R}$ on pairs (a, b): two counters that grow, collide, and *renew*. The machine halts exactly when the orbit enters a small **fatal set** — when the gap between the counters lands in $\{1, 3, 5\}$.

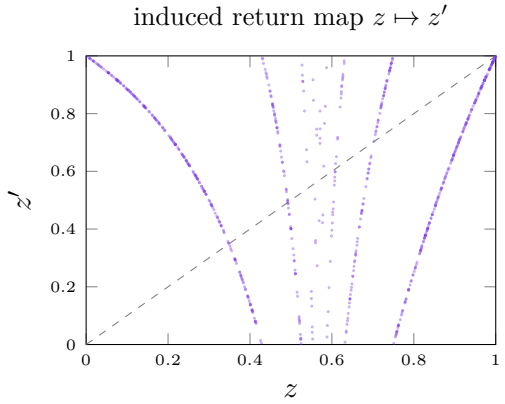
Deciding whether *one* given seed halts is Collatz-hard. So we asked the softer question Tao answered for Collatz: does *almost every* seed run forever?

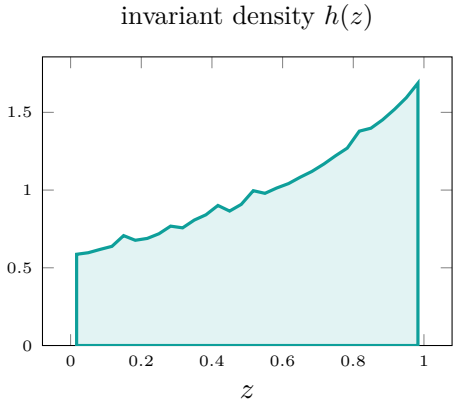
2 From a machine to a Gauss map

The first-return map of $\mathbf{R}$ to a natural section is, in the right coordinate, a **countable full-branch piecewise-Möbius Markov map** G of the interval — a cousin of the Gauss map from continued fractions. Its branch boundaries are the exact fractions

$$\rho_{k+1} = \frac{1}{3\rho_k + 1}, \qquad \rho_k = 1,\ \tfrac{1}{4},\ \tfrac{4}{7},\ \tfrac{7}{19}, \ldots \ \to\ \tfrac{\sqrt{13}-1}{6}.$$

Each branch expands; the whole map has a spectral gap. The left plot shows the return map $z \mapsto z'$ (the expanding Möbius branches); the right shows its invariant density — smooth, bounded, the fingerprint of a well-behaved chaotic map.

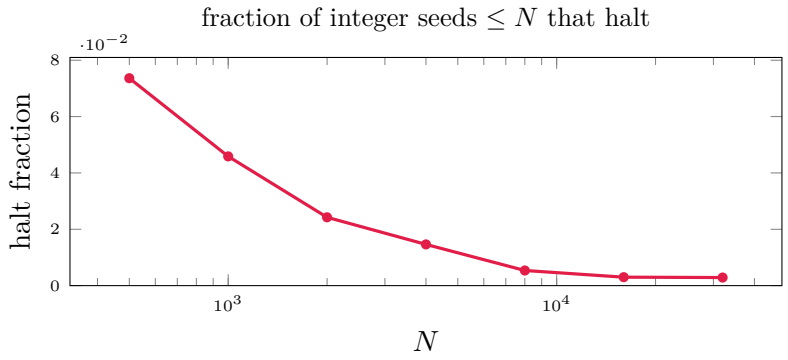




3 Two theorems, one proved

Measure form (proved). Lebesgue-almost every real seed never halts. This is a complete theorem: the Gauss-type map has a unique smooth invariant measure with exponential mixing (Rychlik), the size $a + b$ strictly increases every step, and the halting hazard at scale S is $\le 6h(0)/S$, geometrically summable.

Density form (conditional). The natural density of halting *integer* seeds is zero. Directly confirmed by counting — the halting fraction falls like C/N:



But turning that count into a theorem needs the integer orbits to *equidistribute* as well as the measure does — and that is where harmonic analysis enters. (Companion note: *The Twisted Transfer Operator.*)

Every figure is computed from the exact machine. The measure theorem is complete and axiom-checked in Lean at the closure level; the integer-density statement is reduced to a single, precisely stated analytic lemma and remains conditional. No specific seed is decided.